

Exercise 13

Question 1

Zonal equation discretized. & Meridional equation discretized

Given $\frac{\partial u_E}{\partial t} - f v_E = A_v \frac{\partial^2 u_E}{\partial z^2}$ & $\frac{\partial v_E}{\partial t} = f u_E = A_v \frac{\partial^2 v_E}{\partial z^2}$

Defining the grid: $U_i^n = U_E(z_i, t_n)$

$$V_i^n = V_E(z_i, t_n)$$

Where: $z = i \Delta z$

$$t_n = n \Delta t$$

Time Derivative (forward differences)

$$\frac{\partial u_E}{\partial t} \approx \frac{U_i^{n+1} - U_i^n}{\Delta t} \quad \& \quad \frac{\partial v_E}{\partial t} = \frac{V_i^{n+1} - V_i^n}{\Delta t}$$

Second Vertical Derivative (central difference)

$$\frac{\partial^2 u_E}{\partial z^2} = \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{\Delta z^2}$$

$$\frac{\partial v_E}{\partial z^2} = \frac{V_{i+1}^n - 2V_i^n + V_{i-1}^n}{\Delta z^2}$$

Substitute into equation 42 (zonal equation)

$$\frac{\partial u_E}{\partial t} - f v_E = A_v \frac{\partial^2 u_E}{\partial z^2}$$

$$\frac{U_i^{n+1} - U_i^n}{\Delta t} - f v_i^n = A_v \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{\Delta z^2}$$

$$U_i^{n+1} - U_i^n = f \Delta t V_i^n + \frac{A_v \Delta t}{\Delta z^2} (U_{i+1}^n - 2U_i^n + U_{i-1}^n)$$

$$\therefore U_i^{n+1} = U_i^n + f \Delta t V_i^n + \frac{A_v \Delta t}{\Delta z^2} (U_{i+1}^n - 2U_i^n + U_{i-1}^n)$$

Substitute meridional equation into equation 42

$$\frac{\partial V_E}{\partial t} + f u_E = A_v \frac{\partial^2 V_E}{\partial z^2}$$

$$\frac{V_i^{n+1} - V_i^n}{\Delta t} + f u_i^n = \frac{A_v}{\Delta z^2} (V_{i+1}^n - 2V_i^n + V_{i-1}^n)$$

$$V_i^{n+1} - V_i^n = -f \Delta t u_i^n + \frac{A_v \Delta t}{\Delta z^2} (V_{i+1}^n - 2V_i^n + V_{i-1}^n)$$

$$V_i^{n+1} = V_i^n - f \Delta t u_i^n + \frac{A_v \Delta t}{\Delta z^2} (V_{i+1}^n - 2V_i^n + V_{i-1}^n)$$

Question 2 (Non-dimensional Courant number)

The diffusion term coefficient

$$\frac{A_v \Delta t}{\Delta z^2}$$

The diffusion Courant number

$$C_d = \frac{A_v \Delta t}{\Delta z^2}$$

An explicit FTCS scheme, stability is required

$$C_d \leq \frac{1}{2}$$

$$\therefore \frac{A_v \Delta t}{\Delta z^2} \leq \frac{1}{2} \quad \text{or} \quad \Delta t \leq \frac{\Delta z^2}{2A_v} \quad \text{or} \quad \mu = \frac{A_v \Delta t}{\Delta z^2} \leq \frac{1}{2}$$

Question 3

Using vertical eddy viscosity: $A_v = 10^{-4} \text{ m}^2 \text{ s}^{-1}$

Using the diffusion Courant number from Question 2

$$N = \frac{A_v \Delta t}{\Delta z^2}$$

Apply the stability condition

$$N \leq \frac{1}{2}$$

$$\therefore \frac{A_v \Delta t}{\Delta z^2} \leq \frac{1}{2}$$

$$A_v \Delta t \leq \frac{\Delta z^2}{2}$$

$$\Delta t \leq \frac{\Delta z^2}{2A_v}$$

Using the vertical eddy viscosity

$$\Delta t \leq \frac{\Delta z^2}{2(10^{-4})}$$

$$\Delta t \leq \frac{\Delta z^2}{0.0002}$$

$$\Delta t \leq 5000 \Delta z^2$$

$$\text{Let } \Delta z = 1 \text{ m} \quad \Delta t = 5000 \text{ s}$$

$\therefore$ the diffusion Courant number is $N = 0.25 < 0.5$

the numerical scheme is stable.